

The **algebraic axioms, and what defines them in the Frobenius calculus**

Freyd, *Categories, Algebras* §11-§210 against functorialSemanticsforRelationalTheories.pdf (v14.1)

Lean: Freyd/S2_10.lean(algebra;diag/CB.lean(CartBicat;diag/CB.Algebra).lean(algebraOfCartBicat)

Where the pictures come from: Every picture of a theorem is emitted by ./scripts/diag-export from that theorem's Lean type. Nothing about a theorem is returned here. The two figures marked *transcribed* are the exception: they copy the paper's definition pictures, and definition has no statement to export.

Equation numbering. The numbers below are the 2017 preprint's. The CartBicat fields no longer encode them – a field is named for what it says, $\langle \nabla, \Delta \rangle \vdash \text{true}$ rather than $\text{true} \vdash \langle \nabla, \Delta \rangle$ – and each dotstring carries the preprint number. The repo's copy of the paper is now the published version – Pops J. Math. 2 (6): 30 Oct 2025, doi:10.67905/polygmath.2.6 – which numbers: Translation, published – preprint (34)–(37), (35)–(38), (39)–(39), (37)–(40), (38)–(41), (39)–(42), (40)–(43), (35)–(35), and Example 2.3(c)'s (13)(14)–(11)(12); the preprint's (56) is unnumbered there. Pages Def. 4.1 at 20–21 (was 17), the ∇ notation at 21 (was 19), Lemma 4.2 at 21–22 (was 19), Renumbering the Lean fields is a separate decision and has not been made.

Freyd's cartesian bityrees reciprocation * and intersection $\cap$ as **primitive operations** and lists eight **equational axioms** on them. A cartesian bityree of relations takes neither: every object carries one special Frobenius structure $\Delta, \nabla, ?$, the hom-order ∇ is primitive, and both * and $\cap$ are **defined** from the four dots. Every one of Freyd's eight axioms is then a theorem. The table below is that dictionary, one row per axiom.

Reading the pictures: left to right is composition – the book's diagram order, R S means first R then S – vertical stacking is $\otimes$, a bare wire is an identity, a white box is relation, and a box that is mirrored and tinted is its converse – the mirroring is the paper's own notation, the tint is this note's, because at a glance a chamfer is easy to miss. The label inside is not mirrored; the paper can reverse a single letter and still be read, a whole term like R $\cup$ S cannot. All four dots are solid, as Def. 4.1 draws them. The exporter draws a meet or a converse as a **labelled box**, so a composite under a * appears as a single box reading R ; S ; a $\otimes$ draws as two stacked nodes, and any subterm with no shape in this calculus – an associator, say – degrades to an opaque box. The term columns write the converse *, Freyd's symbol, where the writer writes R† these terms are read against the book throughout, and one symbol per idea beats against the reader to match two. The Lean sources write R* as well. The paper reserves (-) for the colour swap of its §7, but §7 is a worked example in other modes – see the caution at the end of this note – and nothing here formalises it, so in a repo that is R*(Set) throughout there is no clash to protect against.

In a chain table the symbol at the head of each row is a chain of one $\leq$ between axioms – as for a reshaping, $\leq$ for a genuine inequality. Counting them the point of drawing a proof at all – a relation of one $\leq$ proved axioms says the theorem is one law and a lot of bookkeeping.

1. Freyd's primitives are the Frobenius calculus's definitions

Freyd's primitive, and what defines it here	picture
reciprocation $R^\circ := \$21.1$, primitive. Here conv $R := \text{bend } (R \otimes 1) ; \text{cap}$ (diag/CB.lean:223), not a generator. Input arrives bottom left and runs under R; the cup ? ; Δ at the top left creates a pair, one strand feeding R and the other leaving as the output at the top right; the cup ∇ ; Δ at the bottom right annihilates R's output against the input. It is the cup and the cap that reverse the arrow – a wire neatly bumped up over the box and back down is an isotopy of R itself.	
intersection $R \cap S := \$21.1$, primitive. Here the convolution meet $R \cap S := \Delta ; (R \otimes S) ; \nabla$ (diag/CB.Derived.lean:24). Copy the input, run both, merge; the merge forces the two results to agree.	
containment $R \subseteq S := \$21.1$, defined as $R \cap S = R$. Here the poset-enrichment $\leq$ is primitive and $\cap$ is derived, so the two directions are swapped. meet_eq_left_of_le (diag/CB.Derived.lean:196) is the bridge.	no picture – this is the 2-cell itself
maximal morphism $\tau := \$21.5$, and Freyd needs a unitary algebra for it. Here top a b := ! ; ? (diag/CB.Derived.lean:28) is free in Def. 4.1. Freyd.Diag. le_top	

2. The eight axioms, and what proves each

Freyd's axiom, and what supplies it	picture, from the Lean statement
$(R^\circ)^\circ = R := \$21.1$ recip_recip; here the snake , from (41). Freyd.Diag. CartBicat.conv_conv. Bending down and back up straightens out, so bending twice is the identity.	
$(R \cap S)^\circ = R^\circ \cap S^\circ := \21.1 recip_comp; here minoring the picture. Freyd.Diag. CartBicat.conv_conv. Reflecting a chain left to right reverses it; each box comes back flipped.	
$(R \cap S)^\circ = R^\circ \cap S^\circ := \21.1 recip_inter; the same mirroring . Freyd.Diag.conv_inter. Δ and ∇ are each other's mirror image, so reflecting the convolution leaves its shape alone and only turns the boxes round.	
$R \cap R = R := \$21.1$ inter_idem; here (42) and the special law . Freyd.Diag.meet_idem. Copy, run R twice, merge – (42) pushes R back through the copy and Δ ; $\nabla = \text{!}$ closes the bubble.	
$R \cap S = S \cap R := \$21.1$ inter_comm; here commutativity (9) and commutativity (6). Freyd.Diag.meet_comm. The two strands of Δ are interchangeable, and so are those of ∇ .	
$R \cap (S \cap T) = (R \cap S) \cap T := \21.1 inter_assoc; here coassociativity (8) and associativity (5). Freyd.Diag.meet_assoc. Either bracketing builds the same three-way meet tree, and dually the same merge tree.	
semi-distributivity $R (S \cap T) \subseteq R S \cap R T := \21.12 modistrib. §2.136; here (42), the lax comonoid inequation. Freyd.Diag.semIdistrib_of_lax. Running R and then copying is below copying and then running R on each strand.	
modular identity $R \cap S \cap T \subseteq (R \cap S) \cap T := \21.12 modular; the axiom Freyd adjoins: "we adjoin one more containment, the law of modularity". Here (41) and (42), derived. Freyd.Diag.modular_of_frobenius. Copy the shared prefix Δ ; $(R \otimes T)$ off both sides and what is left is $\langle \nabla, \text{slide_conv} \rangle$ (diag/CB.lean:566) – a box slides through a merge at the cost of its converse on the other strand.	

3. The other direction: what each law of Def. 4.1 says in Freyd's language

picture, from the Lean statement	Def. 4.1	Freyd
	$(37) \nabla ; \Delta \leq \text{!}$ half of $\Delta \vdash \nabla$	$\Delta^* \Delta \vdash 1 - \Delta$ is simple §2.13. Not used anywhere in the bridge; see below.
	$(38) \text{!} \leq \Delta ; \nabla$ the other half of $\Delta \vdash \nabla$	$1 \in \Delta^* \Delta - \Delta^*$ is entire §2.13, with (39). ! is a map. This is what makes $\text{!} = \text{!}$; contain every arrow.
	$(39) ? ; \text{!} \leq \text{!}$ half of $\text{!} \vdash ?$	$\text{!}^* \text{!} \vdash 1 - \text{!}$ is simple §2.13. Also unused in the bridge.
	$(40) \text{!} \leq \text{!} ; ?$ the other half of $\text{!} \vdash ?$	$\text{!} \in \text{!}^* \text{!} - \text{!}^*$ is entire §2.13, with (39). ! is a map. This is what makes $\text{!} = \text{!}$; contain every arrow.
	(41) the Frobenius law as an axiom: $\text{!} \otimes \Delta$ and $\nabla \otimes \text{!}$ draw as stacked nodes; only the associator α^{-1} stays an opaque box; having no shape in this calculus.	the modular identity §2.112 – and reciprocation §2.11 itself, since the cup and cap only satisfy the snake because of (41).
	the special law not an axiom: derived from (38) and (40). Freyd.Diag. CartBicat.special	$\Delta^* \Delta^* = 1 - \Delta^*$ is injective , so by §2.141 at $f = g = \Delta$ it is a monic map . In Rel(Set): copying is total and loses nothing.
	(42) every arrow is a lax comonoid homomorphism	§2.136's $R (S \cap T) \subseteq R S \cap R T$, read at $S := \text{!}^* \text{!}$, $T := \text{!}^* \text{!}^*$ – Freyd's semi-distributivity, product-free.
	(43) every arrow is lax for the counit	§2.152's step "p_a is maximal in (α, λ) , hence $R \cap \beta \subseteq p_a \text{!}^*$ " – that is, $R \subseteq T$.

4. Unitary and pre-tabular – the two conditions the converse needs

Freyd's §2.11 axioms are all a cartesian bityree has to give back. Going the other way needs two further conditions, and they are exactly the two the repo has not formalised. Both definitions below are quoted from Freyd/S2_10.lean, which follows the book.

why these pictures are not pictures of a converse		
not, and that is the thing to get straight before reading them. The converse is drawn with two bends – a cup on the way out, each pair of dots, $?$ and ∇ and $!$ – which is the left-hand figure below. The proof never says an arrow is the converse of R as soon as it satisfies one equation between UNBENT arrows $a \bullet b \rightarrow \mathbb{I}$. To the left, one cap on the right. Bending is a bijection (bend_unbend/unbend_bend), so settling the unbent equation settles the cap never has to be drawn at all. every row below has a cap and no cup, and why an arrow that “is a converse” shows up not as a bent box but as the bottom strand, mirrored.		
the converse, as the paper draws it		the one rule the proof uses, twice

Together they are the whole gap: cartesian bityree of relations $\equiv$ unitary pre-tabular algebra (DiagrammaticAlgebraOfFirstOrderLogic §10, after Carboni–Walters). One half of that equivalence is algebraOfCartBicat and is a theorem in this repo; the other half is the two rows above.

5. Union and residual – what phases 8 and 9 add

$\cup$ and / are past where Freyd's §2.11 list stops, and past what Def. 4.1 can build: they need a **bi-product** on top of the Frobenius structure (diag/Tape.lean, after TapeDiagrams.pdf) and a **second composition** with linear adjoints (diag/F0.lean, after DiagrammaticAlgebraOfFirstOrderLogic.pdf). Both are built, so the bound draws.

The two notations say which layer a picture is in. A union is a **tape**: the rounded wrapper is the second monoidal product, and its $\triangleright$ and $\triangleleft$ open and close a branch a particle takes exactly one of – the same fork-join shape as a meet, on a different product. A residual is a **dashed box**, because it is not a composite of the generators at all: $R / S := R ; * S$ lives in the second composition, and a solid box would claim it was drawable in the first.

statement	picture, from the Lean statement
$R \subseteq S := \$22.2$ in Freyd; here Freyd.Diag.BiProd.le_union_left. union $R \subseteq S := (R, S) ; \{1, 1\}$ (diag/Tape.lean:84): offer both branches, then forget which was taken.	
$R \cup S := S \cup R := \22.2 in Freyd; here Freyd.Diag.BiProd.union_comm. The two branches of a tape are interchangeable, exactly as the two strands of Δ are.	
$\text{bot } U = R := \text{Freyd.Diag.BiProd.bot_union}$. $\text{!} \otimes a = \text{toZero } a ; \text{toZero } b$ (diag/Tape.lean:87), routed through the zero object; the exporter has no shape for it, so it prints as a box.	
$R (S \cap T) = R S \cap R T := \22.2 in Freyd; here Freyd.Diag.BiProd.comp_union; composition distributes over union, which $\cap$ does not. Freyd §2.2's distributivity, the reason a distributive algebra is more than an algebra.	
$(R \cup S)^* = R^* \cup S^* := \22.2 in Freyd; here Freyd.Diag.BiProd.conv_union. The converse is a 2-functor for the tape layer too.	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T) := \22.2 in Freyd; here Freyd.Diag.BiProd.distrib. The box that makes distributiveAlgebraOfF0Cb a distributive algebra (§2.2). The one picture with both layers in $\triangleright$: meets inside off the box on the left, tapes around meet on the right.	
$(R / S) \subseteq R / T := \22.2 in Freyd; here Freyd.Diag.ClosedLinearBicat.residual_comp.le; the residual is a lower bound of the arrows it is the greatest of. Freyd's §2.31 division: R / S is the largest T with $T \subseteq R$, and le_residual_iff (diag/F0.lean:170) is that Galois staying in the full. The dashed frame marks the box as belonging to the second composition. The dashed frame marks the box as belonging to the second composition.	

6. Lemma 4.2 (ii) – the proof, not just the statement

diag-export – proof draws a **proof**. A calc chain elaborates to nested Trans, trans, so its intermediate terms are all recoverable from the proof term: flatten the spine, infer the type of each hole, draw one canvas per step. Nothing is replayed and nothing is retold – the nine rows below are the chain the Lean kernel accepted for Freyd.Diag. CartBicat.conv_comp, which is Lemma 4.2 (ii). (R ; S)* = S* R*.

6.1. First: why these pictures are not pictures of a converse

They are not, and that is the thing to get straight before reading them. The converse is drawn with two bends – a cup on the way in and a cap on the way out, each a pair of dots, ? ; Δ and ∇ ; ! – which is the left-hand figure below. The proof never draws that. conv_unslide says an arrow is the converse of R as soon as it satisfies one equation between UNBENT arrows $a \otimes b \rightarrow \text{!}$. Two strands in on the left, one cap on the right. Bending is a bijection (bend_unbend/unbend_bend), so setting the unbent equation settles the bent one, and the cap never has to be drawn at all.

That is why every row below has a cap and no cup, and why an arrow that "is a converse" shows up not as a bent box but as a box sitting on the **bottom** strand, mirrored.

the converse, as the paper draws it	the one rule the proof uses, twice
R^* is R with both wires bent round: in at the bottom left, cap at the top left, cap at the bottom right. Two dots at each bend, as functorialSemanticsforRelationalTheories.pdf does it – a cup is ? then Δ , a cap is ∇ then ! . Transcribed by hand; a definition has no statement to export.	conv_slide (diag/CB.lean:449) a box on the BOTTOM strand facing a cap is the same as that box on the TOP strand, upright. Sliding round the bend is what turns R* into R. This is the entire content of Lemma 4.2 (ii). The other seven steps are bookkeeping.

6.2. The chain

One row per term, not per step; the right-hand side of every step is the left-hand side of the next, so drawing steps would draw every picture twice. The first column is the term as the Lean has it, fully bracketed, with a binding tighter than – ; it comes off the same way as the picture, so the two cannot drift apart. Watch which **strand** each box is on: bottom means mirrored, top means upright.

Lemma 4.2 (ii), (R ; S)* = S* ; R*	picture	the rule that reaches it
term $\text{!} \otimes (S^* ; R^*) ; \text{!}$ (the nine rows below are the chain the Lean kernel accepted for Freyd.Diag. CartBicat.conv_comp, which is Lemma 4.2 (ii). (R ; S)* = S* R*.		the start . What conv_unslide asks about S* R* both boxes on the bottom strand, mirrored – that is what "these are converses" looks like once the wires are unbent.
$= \text{!} \otimes S^* ; (\text{!} \otimes S^*) ; \text{cap}$		1. $\otimes$ is a functor , so doing two things to one strand in sequence is the sequence of doing them. Same picture.
$= \text{!} \otimes S^* ; (R \otimes \text{!}) ; \text{cap}$		2. conv_slide . R* is the box nearest the cap, so it slides round the bend: it leaves the bottom strand, arrives on the top, and comes back upright as R. The first of the two steps that prove anything.
$= (\text{!} \otimes S^* ; R \otimes \text{!}) ; \text{cap}$		3. simp only [Cat.assoc]. Re-bracketing – stable in the term column, invisible in the picture. Same picture.
$= R \otimes S^* ; \text{cap}$		4. tensHom_split* – interchange. R is on the top strand and S* on the bottom, so running them in sequence and running them side by side are the same thing. Shows as a re-queuing only because the exporter lays a diagram out left to right.
$= (R \otimes \text{!} ; \text{!} \otimes S^*) ; \text{cap}$		5. tensHom_split* – the same law, the other way. Split them apart again. R first, so that S* is now the box facing the cap. Which is what the next real step needs.
$= R \otimes \text{!} ; (\text{!} \otimes S^*) ; \text{cap}$		6. simp only [Cat.assoc]. Re-bracketing. Same picture.
$= R \otimes \text{!} ; (S \otimes \text{!}) ; \text{cap}$		7. conv_slide , on S this time. S* slides round the bend and arrives upright on the top strand. Both boxes are now on top, both upright, in the order R then S. The second and last real step.
$= (R \otimes \text{!} ; S \otimes \text{!}) ; \text{cap}$		8. simp only [Cat.assoc]. Re-bracketing. Same picture.
$= (R ; S) \otimes \text{!} ; \text{cap}$		9. tensHom_split* – interchange once more: the two top-strand boxes merge into one. This is R ; S unbent, so conv_unslide reads off S* R* $\otimes$ (R ; S)*. Done.

What the **chain measures**. Read down the pictures and most consecutive pairs are the same drawing – steps 1, 3, 6, 8 and 9 change nothing at all, and step 4 changes only spacing – while the term column builds them changes at every row. Those are associativity, the unit laws and interchange – precisely the laws a string diagram has into their geometry rather than as rewires. Seven of these nine steps are bureaucratic picture language does not charge for, and the theorem is two applications of conv_slide. How this differs from the paper's proof. The paper proves (ii) by inserting a snake in the middle of R* ; S and reading the two halves off as S* and R* – three pictures, all bent. The Lean never bends. Same theorem, different route, and the export shows the route the kernel accepted. Drawn by hand it would have shown the paper's three pictures and quietly implied those were what the repo had checked.

7. The modular law – proved, drawn

Freyd adjoins the modular identity: an algebra has no structure to derive it from, so it goes in as an axiom. Here it is a theorem, and this is its proof – Freyd.Diag.modular_of_frobenius (diag/CB.Algebra.lean:48), four terms, of which exactly one step is an inequality.

the modular law, $R \cap S \cap T \subseteq (R \cap S) \cap T$	picture	the rule that reaches it
$(R ; S) \cap T$		Factor out the shared prefix . (R S) $\otimes$ T is (R $\otimes$ T) ; (S $\otimes$ T), so even before the S is common to both sides and can be set aside. An equality, and drawn as one.
$\leq (\Delta ; R \otimes T) ; (S \otimes \text{!}) ; \nabla$		$\langle \nabla, \text{slide_conv} \rangle$, under the shared prefix . S slides through the merge and reappears as S* on the other strand. The only inequality in the proof – everything else is folding.
$= R \cap (T ; S^*) ; S$		Red the prefix back in . R $\otimes$ (T S*) is (R $\otimes$ T) ; (T $\otimes$ S*), and the copy-meet pair reads as a meet again. (R $\cap$ T S*) $\subseteq$ S, which is the goal.

7.1. Inside the middle step

$\langle \nabla, \text{slide_conv} \rangle$ (diag/CB.lean:560) has where the mathematics is, and it mentions only $S \rightarrow R$ and T sit untouched in the prefix throughout. Draw on its own it says one arrow slides through a merge:

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S occurs once on the left and twice on the right. That is the tell: the lax cup inequation (42) is the only law in Def. 4.1 that may duplicate a box, so it has to be the inequality step – and its own chain shows it is the only one.

$\langle \nabla, \text{slide_conv} \rangle, (S \otimes \text{!}) ; \nabla \leq (\text{!} \otimes S^*) ; \nabla ; S$	picture	the rule that reaches it
term $S \otimes \text{!} ; \nabla$		the start : (S $\otimes$ T) $\leq \nabla$.
$= (S ; \Delta) \otimes \text{!} ; (\alpha ; (\text{!} \otimes \text{cap}) ; \nabla) ; S$		$\langle \nabla, \text{of_cap} \rangle$, backwards . The merge is a return from a copy and a cap, which puts the term into the shape (S ; Δ) $\otimes$ T that (42) applies to.
$\leq (\Delta ; (S \otimes S) \otimes \text{!}) ; (\alpha ; (\text{!} \otimes \text{cap}) ; \rho) ; S$		(42), lax Δ . Δ becomes $(\Delta ; (S \otimes S))$; the box is copied. The whole of the modular law, in one picture.
$= \text{!} \otimes S^* ; (\nabla ; S)$		$\langle \text{cap_tens_v} \rangle$ and conv_slide . Reshaping back, and the surviving duplicate slides round the cap to become S* – the same rule Lemma 4.2 (ii) is made of.

What the **associators cost**, α and ρ have no shape in this calculus and print as opaque boxes, so the two middle terms of the inner chain are less legible than its ends. That is the honest price of drawing what the Lean says: the reshaping steps really are that fiddly, and a hand-drawn version would have hidden them by working up to coherence.

8. Maps: SV, TOT, INJ, SUR, and the two inequations that always hold

Def. 4.1 clause 4 says every arrow is lax for Δ and for ! – inequations (42) and (43). Applying Lemma 4.2 (ii) and (iv) to them gives (44) and (45), the same statements about ∇ and $?$. Those four hold for every arrow. Their four REVERSES do not, and each one failing or holding is a property of the arrow: that is (SV), (TOT), (INJ), (SUR).

holds for every arrow	holds iff the arrow is ...
(42) $R ; \Delta \leq \Delta ; (R \otimes R)$	(SV) – single valued
(43) $R ; \text{!} \leq \text{!} ; R$	(TOT) – total
(44) $\nabla ; R \leq (R \otimes R) ; \nabla$ – the converse of (42). Not in the Lean; see below.	(INJ) – injective
(45) $? ; R \leq ?$ – the converse of (43). Not in the Lean; see below.	(SUR) – surjective

The paper gives the same four properties a second time, as ADJOINT conditions – (46)–(49) – and Lemma 4.4 says the two presentations agree. $\text{diag/CB.Derived.lean}$ takes the adjoint forms as the definitions, because they are literally Freyd's Simple and Entire (§2.13), so a diagram map and a book map are one predicate with no translation layer.

property	adjoint form, the definition	Freyd
single valued Freyd.Diag.SingleValued		Simple $R := R^* R \subseteq \text{!}$ §2.13
total Freyd.Diag.Total		Entire $R := \text{!} \subseteq R R^*$ §2.15. With single valued, R is a map.
injective Freyd.Diag.Injective		$R^* R \subseteq \text{!}$ §2.141 at $f = g = R$. A monic pair, so a monic map.
surjective Freyd.Diag.Surjective		$\text{!} \subseteq R^* R$ §2.147 "g is a cover iff $\text{!} \subseteq g^* g$ ", that is, iff g^* is entire.

Why (44) and (45) have **no picture**. Both are converse-images, so drawing them from a Lean statement needs that statement to get the chain does not. Getting it needs $\Delta^* = \nabla, \text{!}^* = ?$ and Lemma 4.2 (iii) ($f \otimes g^* = f^* \otimes g^*$) – and the first of those is a Frobenius computation at a COMPOSITE object, which is the clause of Carboni & Walters' definition that Def. 4.1 at printed omits (diag/CB.lean's header records this). It is the same gap that leaves Lemma 4.4 unproved in the repo. The four reverse inequations opposite are unaffected: they are recorded under the paper's own labels in diag/CB.Derived.lean, and nothing downstream depends on Lemma 4.4.